What goes with happiness? A survey analysis of US adults, 2024

Sample project built with public data. Data: General Social Survey (GSS) 2024, NORC at the University of Chicago. Analysis and report by Marlon Jiménez Valencia.

Question. Which personal circumstances are associated with saying you are "very happy"? We look at marital status, self-rated health and satisfaction with one's financial situation, and then estimate all of them together, with age, sex, education and religious attendance, in a logistic regression.

1. Data and method

- **Survey.** The GSS is a nationally representative, face-to-face and web survey of US adults run since 1972. The 2024 file has 3,986 respondents. Variable definitions come from the official GSS 2024 codebook (see `docs/variables.md`).
- **Weights and design.** All percentages use the weight `WTSSNRPS_AS`, which NORC recommends for analysing all 2024 cases together (including the AmeriSpeak oversample). Confidence intervals and the regression standard errors take into account the stratified, clustered design (`VSTRAT`, `VPSU`), so they are wider than simple random-sample formulas would give.
- **Missing data.** "Don't know", "no answer" and "skipped" are treated as missing; each analysis uses the respondents with valid answers for its variables (complete cases).

Respondents: **3,986** · valid answers to the happiness question: **3,954** · variance strata: 68 · primary sampling units: 137.

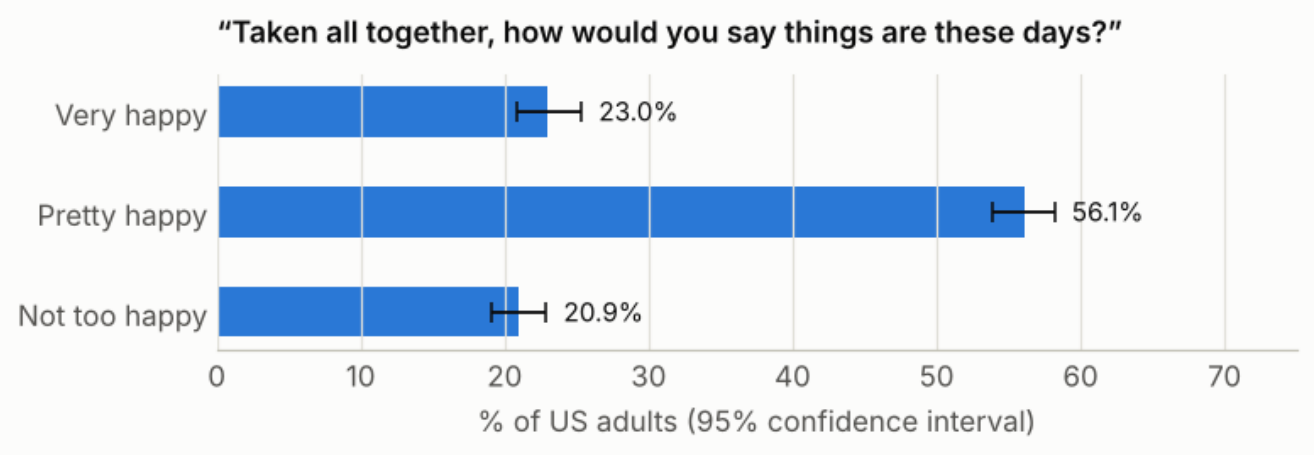
2. Who is in the sample

Weighted percentages describe the US adult population; *n* is the number of respondents (unweighted).

Variable	Group	Weighted %	n
Sex	Female	51.0	2,169
	Male	49.0	1,787
Age	18-34	30.9	1,074
	35-49	25.4	1,057
	50-64	22.3	857
	65+	21.5	871

Education	Bachelor's or higher	33.6	1,440
	No bachelor's degree	66.4	2,539
Marital status	Married	51.1	1,659
	Never married	30.4	1,334
	Divorced / separated	13.2	709
	Widowed	5.3	269
Self-rated health	Excellent	18.1	665
	Good	54.2	2,151
	Fair	23.5	975
	Poor	4.2	173
Financial satisfaction	Satisfied	22.1	887
	More or less satisfied	47.4	1,810
	Not satisfied at all	30.5	1,267
Religious attendance	Less than monthly	63.9	2,574
	Monthly or more	36.1	1,369

3. How happy are Americans?



About **23.0%** of US adults say they are *very happy* (95% CI 20.8% to 25.3%); 56% say *pretty happy*. The design effect for this estimate is 2.8: the clustered sample makes the interval about 1.7 times as wide as a simple random sample of the same size would.

4. Cross-tabulations and chi-square tests

Each table shows **row percentages** (weighted): for example, of married adults, what share is very happy. The **chi-square test** checks whether happiness and the other variable are independent (null hypothesis: no association). It is computed on the respondent counts; all expected counts are well above 5, so the test is valid. **Cramér's V** measures the strength of the association (0 = none, about 0.1 = small, 0.3 = medium).

Happiness by marital status (row %, weighted)

happiness marital	Very happy	Pretty happy	Not too happy	n
Married	31.7	56.1	12.3	1,647
Never married	13.4	55.0	31.7	1,325
Divorced / separated	15.3	58.9	25.8	702
Widowed	15.1	53.7	31.2	266

$\chi^2(6) = 211.5$, $p < 0.001$, Cramér's V = 0.16, n = 3,940, smallest expected count = 54.

Happiness by self-rated health (row %, weighted)

happiness health	Very happy	Pretty happy	Not too happy	n
Excellent	40.8	45.4	13.8	663
Good	23.3	59.6	17.1	2,136
Fair	10.2	59.6	30.2	972
Poor	14.7	35.0	50.3	171

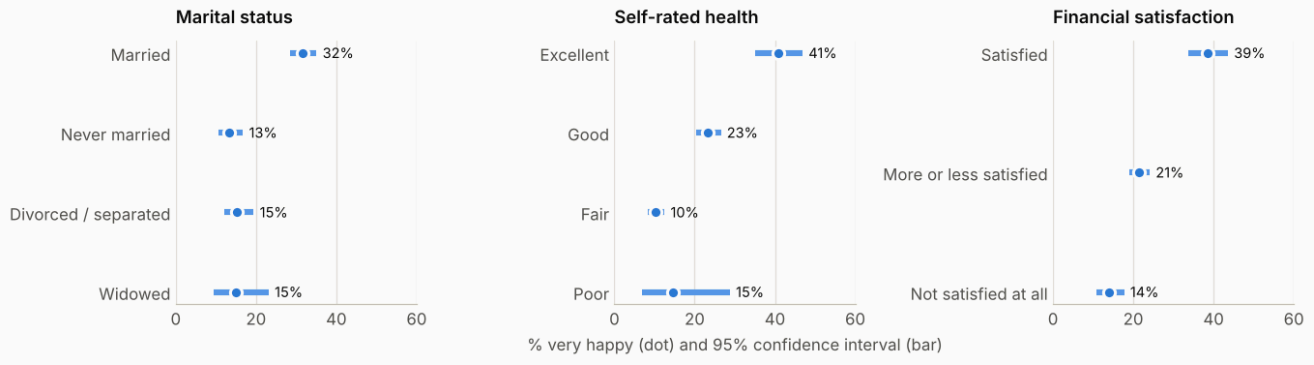
$\chi^2(6) = 338.9$, $p < 0.001$, Cramér's V = 0.21, n = 3,942, smallest expected count = 34.

Happiness by financial satisfaction (row %, weighted)

happiness finances	Very happy	Pretty happy	Not too happy	n
Satisfied	38.6	51.9	9.5	885
More or less satisfied	21.4	62.3	16.3	1,799
Not satisfied at all	13.9	50.2	35.9	1,253

$\chi^2(4) = 396.2$, $p < 0.001$, Cramér's V = 0.22, n = 3,937, smallest expected count = 178.

5. Share who are "very happy", with 95% confidence intervals



- **Marriage:** 32% of married adults are very happy, against 13% of those never married; the confidence intervals do not overlap.
- **Health:** 41% of people in excellent health are very happy, against 15% of those in poor health.

- **Money:** 39% of those satisfied with their finances are very happy, against 14% of those not satisfied at all.

6. Logistic regression: what predicts being "very happy", all else equal?

The cross-tabs look at one factor at a time. Married people are older on average, healthier people tend to have more money, and so on, so the regression estimates the association of each factor **holding the others constant**. Outcome: 1 = very happy, 0 = pretty or not too happy. Reference groups: married, excellent health, satisfied with finances, age 18-34, male, no bachelor's degree, attends religious services less than monthly.

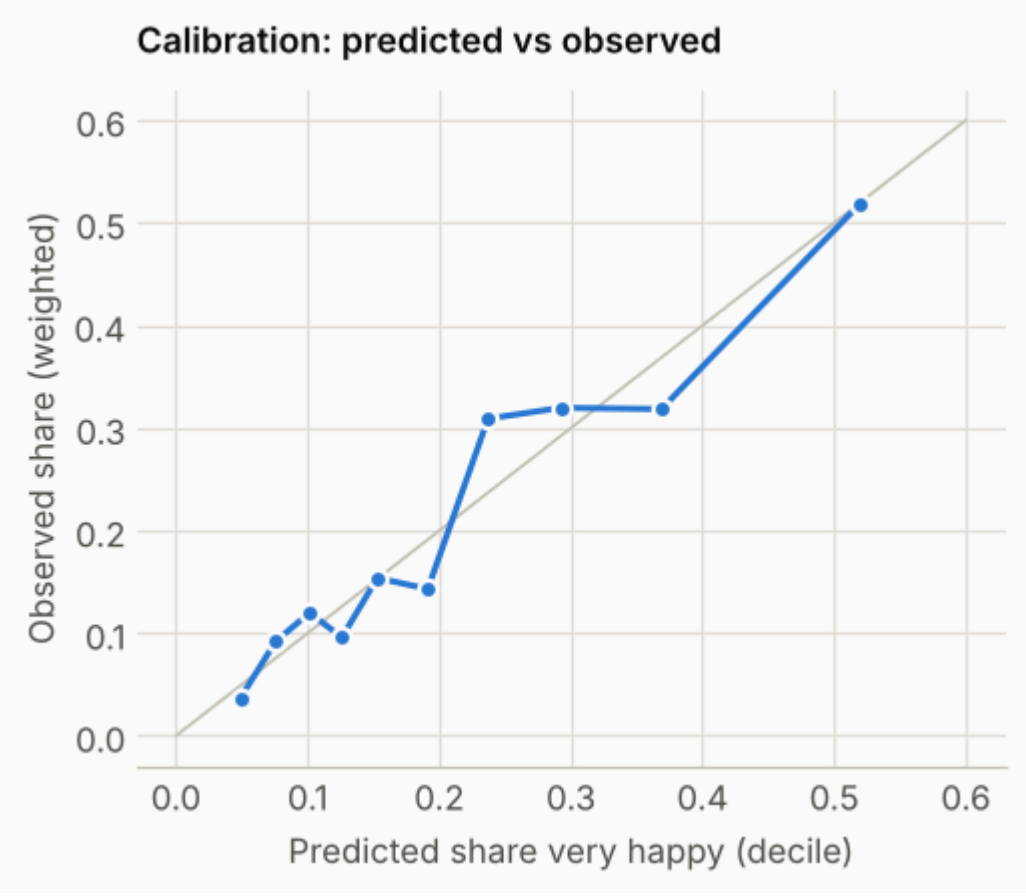
Estimation: weighted logistic regression (survey weight `WTSSNRPS_AS`) with standard errors clustered by primary sampling unit, a standard approximation to design-based ("svy") logistic regression.

	Odds ratio	95% CI low	95% CI high	p-value	Change in probability (pp)
Divorced / separated (vs married)	0.46	0.33	0.64	< 0.001	-12.5
Never married (vs married)	0.35	0.24	0.50	< 0.001	-16.0
Widowed (vs married)	0.39	0.23	0.65	< 0.001	-14.8
Fair health (vs excellent)	0.20	0.14	0.30	< 0.001	-25.5
Good health (vs excellent)	0.46	0.34	0.62	< 0.001	-14.8
Poor health (vs excellent)	0.28	0.12	0.63	0.002	-21.9
Finances: more or less satisfied (vs satisfied)	0.53	0.40	0.69	< 0.001	-11.3
Finances: not satisfied at all (vs satisfied)	0.38	0.27	0.53	< 0.001	-15.9
Age 35-49 (vs 18-34)	0.94	0.66	1.32	0.709	-1.0
Age 50-64 (vs 18-34)	0.73	0.52	1.04	0.080	-4.6
Age 65+ (vs 18-34)	0.95	0.62	1.46	0.815	-0.8
Female (vs male)	1.11	0.87	1.41	0.406	+1.6
Bachelor's degree or higher	0.73	0.56	0.94	0.015	-4.8
Religious services monthly or more	1.37	1.07	1.74	0.011	+4.9

n = 3,760 respondents with complete data (752 very happy). McFadden pseudo-R^2 = 0.115; area under the ROC curve = 0.73.



Checking the model's assumptions



- **Independent observations.** Respondents are clustered by sampling unit; this is handled by the clustered standard errors (137 clusters).
- **Enough events.** 54 events per predictor (rule of thumb: at least 10).

- **No harmful multicollinearity.** Highest variance inflation factor = 2.02 (problems usually start above 5).
- **Linearity of the logit in age.** Age enters as four groups, so no straight-line assumption is made.
- **Influential observations.** Largest Cook's distance = 0.0076, about 7 times the common warning level of 4/n = 0.0011; 270 cases exceed that level, but all distances are far below 1, so no single respondent drives the results.
- **Fit.** Across deciles of predicted probability, observed and predicted shares differ by at most 7.2 percentage points (plot). The Hosmer-Lemeshow test on the unweighted fit gives $\chi^2(8) = 16.9$, $p = 0.03$: it flags some lack of fit. With almost 4,000 respondents the test detects even modest deviations, so read it together with the plot: the model ranks respondents reasonably well (AUC 0.73), but its predicted probabilities can be off by a few points in some groups and should be read as approximate. Pseudo-R^2 0.115 is typical for attitudes: happiness depends on much more than these variables.

7. What this means, in plain language

- **Marriage matters even after accounting for health, money and age.** Compared with otherwise similar married people, the never-married are about **16 percentage points** less likely to say they are very happy (odds ratio 0.35); the divorced or separated about **13 points** less.
- **Health shows the largest gaps.** Compared with people in excellent health, those in fair health are about **26 points** less likely to be very happy and those in poor health about **22 points** less (odds ratios 0.20 and 0.28; the poor-health group is small, so its estimate is less precise).
- **Feeling financially secure matters too.** Those not satisfied at all with their finances are about **16 points** less likely to be very happy than those who are satisfied.
- **Smaller associations.** Attending religious services at least monthly goes with a +5-point difference ($p = 0.011$). Once health and financial satisfaction are held constant, a bachelor's degree goes with a -5-point difference ($p = 0.015$): education seems to matter mainly through the other factors. Sex ($p = 0.406$) and age group (all $p \geq 0.08$) show no clear difference.
- These are **associations, not causes**: happier people may also be more likely to marry or to rate their health as good.

8. Limitations

- **Cross-sectional data:** one point in time, so the direction of the relationships cannot be established.
- **Self-reported answers:** happiness, health and financial satisfaction are the respondent's own assessment.
- **Mode of interview:** in 2024 most respondents answered on the web and the rest face-to-face or by phone; answers to sensitive questions can differ by mode.
- **Complete-case analysis:** respondents with a missing answer in any model variable are left out of the regression; results could change slightly with imputation.
- **Design approximation:** the regression uses weights plus cluster-robust errors, a close approximation to full design-based estimation; the cross-tab chi-square tests use

respondent counts.

Source. Davern, M., Bautista, R., Freese, J., Herd, P. and Morgan, S. L. *General Social Survey 1972-2024* [machine-readable data file], 2024 Release 3a. NORC at the University of Chicago, with principal funding from the National Science Foundation. Data downloaded from gss.norc.org; used under the GSS terms (free use with citation).